



United International University

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Department of Electrical and Electronic Engineering

Analog Integrated Circuits (EEE 4117 (B))

Project Report

Design and Analysis of a Two-Stage Op-amp Using TSMC 45nm PDK

Submitted to:

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Abstract

This project presents the design and analysis of a two-stage operational amplifier (Op-amp) using the TSMC 45nm Process Design Kit (PDK) in the Cadence design environment. The Op-amp consists of a differential amplifier as the first stage and a common-source amplifier as the second stage, powered by a 1.2 V supply with a single current source. The design targets a voltage gain of 55–70 dB, phase margin of 60–70 degrees, bandwidth of 50–60 MHz, input common-mode range (ICMR) of 0.3–0.6 V (ICMR-) and 0.9–1.0 V (ICMR+), slew rate of 5–20 V/ μ s, and an output load capacitance of 1–5 pF, while minimizing DC power consumption. Comprehensive simulations, including DC, AC, transient, and noise analyses, were performed to validate the design. This report details the design methodology, transistor sizing, simulation results, and performance analysis.

1 Introduction

This project involves the design of a two-stage operational amplifier (Op-amp) using the TSMC 45nm PDK in the Cadence design environment. The Op-amp comprises a differential amplifier as the first stage and a common-source amplifier as the second stage, powered by a 1.2 V supply with a single current source. The design targets specific performance metrics, including voltage gain, phase margin, bandwidth, input common-mode range (ICMR), slew rate, and output load capacitance, while minimizing DC power consumption. Comprehensive simulations, including DC, AC, transient, and noise analyses, are performed to validate the design. This report details the design methodology, transistor sizing, simulation results, and analysis, as per the project requirements.

2 Design Methodology

2.1 Circuit Topology

The two-stage Op-amp consists of a differential amplifier with an active load as the first stage, followed by a common-source amplifier with a current source load as the second stage. A compensation capacitor is included to achieve the desired phase margin. The circuit is biased using a 1.2 V supply and a single current source, with transistors selected from the TSMC 45nm PDK library (1.2 V transistors).

2.2 Design Specifications

The target specifications are:

- Voltage gain: 55–70 dB
- Phase margin: 60–70 degrees
- Bandwidth: 50–60 MHz
- ICMR-: 0.3–0.6 V, ICMR+: 0.9–1.0 V
- Output load capacitance: 1–5 pF

- Slew rate: 5–20 V/ μ s

2.3 Hand Calculation

The aspect ratios (W/L) of the transistors were calculated based on the design specifications and the TSMC 45nm PDK parameters.

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Design specification :

voltage gain = 55dB

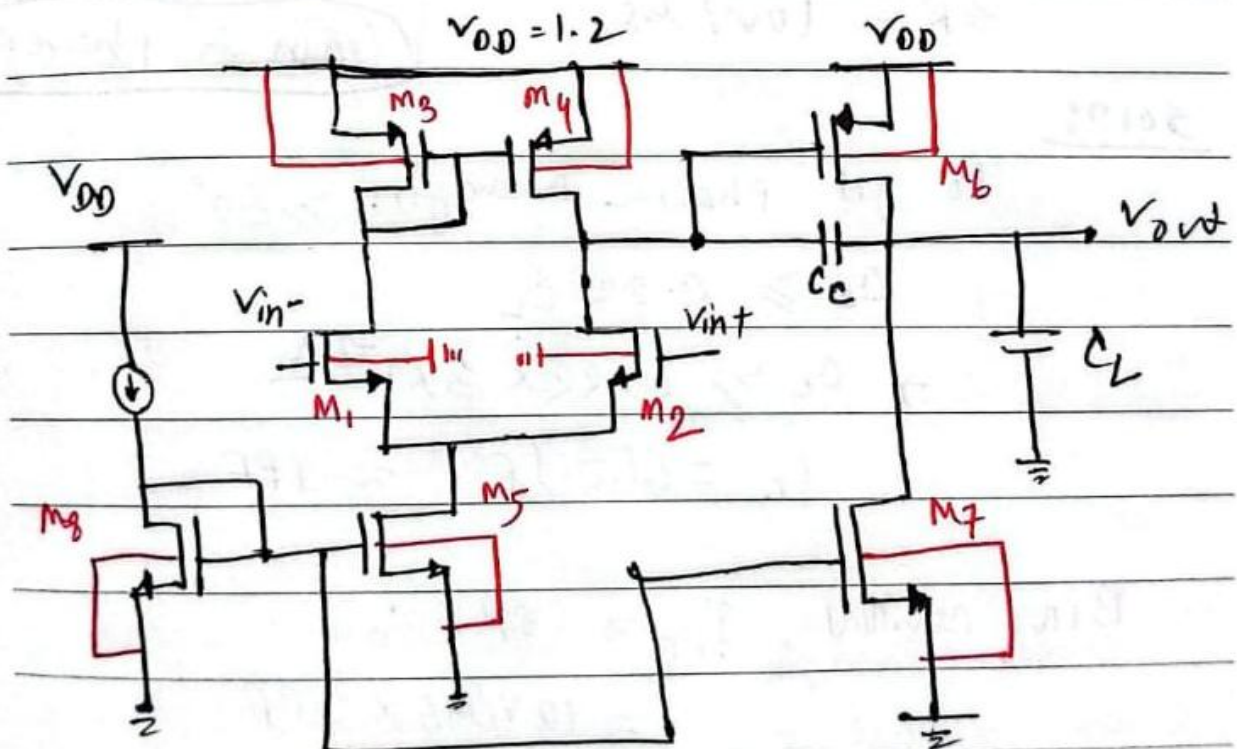
phase margin = 60°

Bandwidth = 54 MHz

Supply voltage = 1.2

ICMR Range = 510mV to 1V

TSMC 45 nm Technology PDK



Single stage Opamp :

$$V_{DD} = 1.2$$

$$\text{Voltage Gain} = 55 \text{ dB}$$

$$\text{Phase Margin} = 60^\circ$$

$$I_{CMR-} = 0.4$$

$$I_{CMR+} = 1 \text{ V}$$

$$C_L = 3 \text{ pF}$$

$$SR = 10 \text{ V/}\mu\text{s}$$

$$GBW = 55 \text{ MHz}$$

Transient Analysis

input $\rightarrow$ sinusoidal signal
output $\rightarrow$ amplitude gain $\times$ phase shift

AC analysis $\rightarrow$ frequency sweep $\rightarrow$ output $\rightarrow$ gain & phase plot

Gain & Phase Plot

Soln:

To get phase margin $> 60^\circ$

$$C_c \geq 0.22 C_L$$

$$\Rightarrow C_c \geq 0.22 \times 3 \times 10^{-12}$$

$$C_c = 660 \text{ fF} \approx 1 \text{ pF}$$

$$\text{Bias current, } I_{D5} = SR \times C_c$$

$$= 10 \text{ V/}\mu\text{s} \times 1 \text{ pF}$$

$$= 10 \text{ nA}$$

we should take a little bit higher from the calculated value.

$$\text{Let, } I_{D5} = 15 \text{ nA}$$

Design M_1 & M_2 :

We know, $g_m = GBW \times C_c$

$$\begin{aligned} \therefore g_{m_{1,2}} &= GBW \times C_c = 2\pi f \times C_c \\ &= 55 \times 10^6 \times 2\pi \times 1 \times 10^{-12} \\ &= 345.5 \text{ MS} \end{aligned}$$

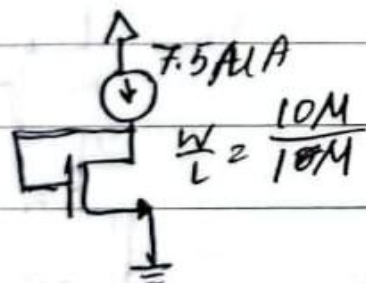
Now,

$$g_{m_{1,2}} = \sqrt{2 \mu_n C_{ox} \left(\frac{W}{L}\right)_{1,2} I_{D_{1,2}}}$$

for this, we need to find out the β_{eff} value.

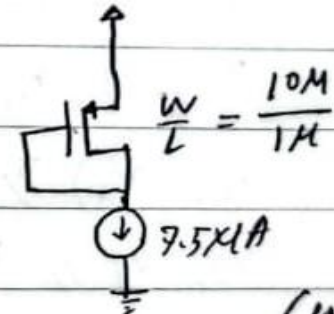
$$\boxed{\beta_{eff} = \mu_n C_{ox} \left(\frac{W}{L}\right)}$$

For NMOS



using the maximum $\left(\frac{W}{L}\right)$ ratio

For PMOS



$$\begin{aligned} \beta_{eff} &= \mu_n C_{ox} \left(\frac{W}{L}\right) \\ \Rightarrow 3.37196 \text{ m} &= \mu_n C_{ox} \left(\frac{10\text{M}}{1\text{M}}\right) \\ \Rightarrow \mu_n C_{ox} &= \frac{3.37196 \text{ m}}{10} \\ &= 337.19 \text{ M} \end{aligned}$$

$$\begin{aligned} \beta_{eff} &= \mu_p C_{ox} \left(\frac{W}{L}\right) \\ \Rightarrow 2.30796 \text{ m} &= \mu_p C_{ox} \left(\frac{10\text{M}}{1\text{M}}\right) \\ \Rightarrow \mu_p C_{ox} &= 230.796 \text{ M} \\ \mu_p C_{ox} &\approx 300 \text{ MA/V}^2 \end{aligned}$$

So, $\mu_n C_{ox} \approx 350 \text{ MA/V}^2$
 $|V_{th,n}| = 400 \text{ mV}$

$|V_{th,p}| = 335 \text{ mV}$

Now, $g_{m1,2} = 345.5 \mu S$

$$g_{m1,2} = \sqrt{2 \mu_n C_{ox} \left(\frac{W}{L}\right)_{1,2} I_{D1,2}}$$

$$\Rightarrow \left(\frac{W}{L}\right)_{1,2} = \frac{g_{m1,2}^2}{2 I_{D1,2} \mu_n C_{ox}} = \frac{(345.5 \times 10^{-6})^2}{2 \times (7.5 \times 10^{-6}) \times 400 \times 10^{-8}}$$

$$\therefore \left(\frac{W}{L}\right)_{1,2} = 22.73$$

$\left(\frac{W}{L}\right)_{3,4}$ Related to $ICMR+$:

$$M_3, M_4 \rightarrow ICMR+ = 1V$$

For Saturation:

$$V_{DS3} \gg V_{GS3} - V_{TH,N}$$

$$\Rightarrow (V_{D1} - V_1) \gg V_{GS3} - V_{TH,N}$$

$$\Rightarrow V_{D1} - V_1 \gg V_{in,CM} - V_1 - V_{TH,N}$$

$$\Rightarrow V_{D1} \gg V_{in,CM} - V_{TH,N}$$

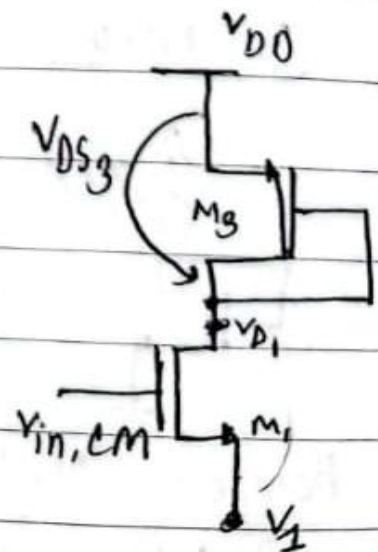
$$\Rightarrow V_{D1} \gg 1 - 0.4$$

$$\therefore V_{D1} = 0.6 = 600mV$$

Now,

$$V_{DS3} = V_{GS3} \quad \& \quad \boxed{V_{SD3} = V_{SG3}}$$

$$\text{So, } V_{SD3} = V_{DD} - V_{D1} = 1.2 - 0.6 = 0.6V$$



Now,

$$I_{D3,1} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L}\right)_3 (V_{SD3} - |V_{Th,p}|)^2$$

$$\Rightarrow 7.5 \times 10^{-6} = \frac{1}{2} \times 300 \times 10^{-6} \left(\frac{W}{L}\right)_{3,1} (V_{SD3} - |V_{Th,p}|)^2$$

$$\Rightarrow \left(\frac{W}{L}\right)_{3,1} = \frac{7.5 \times 10^{-6}}{\frac{1}{2} \times 300 \times 10^{-6} (0.6 - 0.335)^2}$$

$$= 0.7119$$

$$\therefore \left(\frac{W}{L}\right)_{3,1} = 0.7119 \approx 1$$

$M_5, M_8 \rightarrow I_{CMR-1} = 0.4V = V_{in,cm}$

Here, $V_{in,cm} = V_{GS1} + V_{DS5}$

$\Rightarrow V_{in,cm} = V_{GS1} + V_{DS5}$

$\Rightarrow V_{DS5} = V_{in,cm} - V_{GS1}$

$V_{DS5} = V_{in,cm} - V_{GS1}$

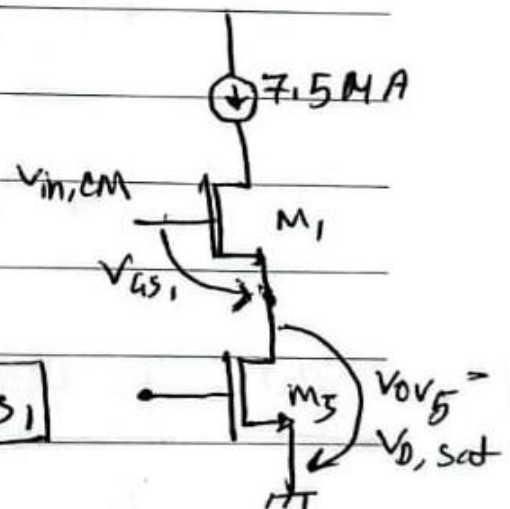
$V_{GS1} = ?$

$I_{D1} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L}\right)_{1,2} (V_{GS1} - V_{Th,n})^2$

$\Rightarrow V_{GS1} = V_{Th,n} + \sqrt{\frac{2 \times I_{D1}}{\mu_n C_{ox} \left(\frac{W}{L}\right)_1}}$

$= 0.4 + \sqrt{\frac{2 \times 7.5 \times 10^{-6}}{350 \times 10^{-6} \times 22.73}}$

$= 0.45V$



$$\begin{aligned} V_{D,sat} &= V_{in,cm} - V_{as} \\ &= 0.4 - 0.45 \\ &= -0.05 \text{ V} \end{aligned}$$

Select $V_{in, min}$ to keep M_5 in Saturation.
 $V_{in, min} = 0.6V$ to keep M_1 & M_5 in Satⁿ mode.

$$\Rightarrow \left(\frac{W}{L}\right)_{5,8} = \frac{2 \times 25 \times 15 \times 10^{-6}}{350 \times 10^{-6} \times (V_{DS5})^2} \quad \rightarrow (-0.05)$$

$$\Rightarrow \left(\frac{W}{L}\right)_{5,8} = 34.28$$

2nd Stage Design : design of MC :

For phase margin γ , 60°

$$\begin{aligned} g_{m6} &= 10 \times g_{m1,2} \\ &= 10 \times 345.5 \times 10^{-6} \\ &= 3.455 \text{ m.S} \end{aligned}$$

and $R_c = \frac{1}{g_{m6}} = \frac{1}{3.455 \times 10^{-3}} = 290 \Omega$

$$\left(\frac{w}{L}\right)_b = \frac{I_{D6}}{I_{D4}} \times \left(\frac{w}{L}\right)_4$$

$$\Rightarrow \frac{\left(\frac{w}{L}\right)_b}{\left(\frac{w}{L}\right)_4} = \frac{I_{D6}}{I_{D4}} = \frac{g_{m6}}{g_{m4}}$$

$$\Rightarrow \boxed{\left(\frac{w}{L}\right)_b = \frac{I_{D6}}{I_{D4}} \times \left(\frac{w}{L}\right)_4}$$

$$\therefore \left(\frac{w}{L}\right)_b = \frac{g_{m6}}{g_{m4}} \times \left(\frac{w}{L}\right)_4$$

$$g_{m4} = \sqrt{2 I_{D4} \mu_p C_{ox} \left(\frac{w}{L}\right)_4} = \sqrt{2 \times 7.5 \times 10^{-6} \times 300 \times 10^{-6} \times 0.7119}$$

$$= 56.59 \text{ } \mu\text{S}$$

$$\text{So, } \left(\frac{w}{L}\right)_b = \frac{3.455 \times 10^{-3}}{56.59 \times 10^{-6}} \times (0.7119)$$

$$\therefore \left(\frac{w}{L}\right)_b = 43.45$$

$$\text{Again, } \frac{\left(\frac{w}{L}\right)_b}{\left(\frac{w}{L}\right)_4} = \frac{I_{D6}}{I_{D4}} \Rightarrow I_{D6} = \left(\frac{\left(\frac{w}{L}\right)_b}{\left(\frac{w}{L}\right)_4}\right) \times I_{D4}$$

$$\therefore \boxed{I_{D6} = 457.81 \text{ } \mu\text{A}}$$

$$I_{D6} = I_{D7}$$

Now,

$$\frac{I_{D7}}{I_{D5}} = \frac{\left(\frac{W}{L}\right)_7}{\left(\frac{W}{L}\right)_5}$$

$$\Rightarrow \left(\frac{I_{D7}}{I_{D5}}\right) \times \left(\frac{W}{L}\right)_5 = \left(\frac{W}{L}\right)_7$$

$$\Rightarrow \left(\frac{W}{L}\right)_7 = \left(\frac{457.81 \mu A}{815 \times 10^{-6}}\right) \times 34.28$$

$$= 31046$$

$$\text{So, } \left(\frac{W}{L}\right)_7 = 31046$$

So, Aspect Ratio,

$$\left(\frac{W}{L}\right)_{1,2} = 22.73$$

$$\left(\frac{W}{L}\right)_{3,4} = 0.7119$$

$$\left(\frac{W}{L}\right)_{5,8} = 34.28$$

$$\left(\frac{W}{L}\right)_6 = 43.45$$

$$\left(\frac{W}{L}\right)_7 = 31046$$

Hence, $L_{\min} = 200 \text{ nm}$

$$L \geq 2 L_{\min}$$

$$\Rightarrow L \geq 400 \text{ nm}$$

We can take, $L = 400 \text{ nm}$

$$\therefore, \left(\frac{w}{L}\right)_{1,2} = 22.73$$

$$\Rightarrow w_{1,2} = 22.73 \times 400 \text{ nm} \quad [L = 400 \text{ nm}]$$

$$= 9.09 \mu\text{m}$$

$$\therefore w_{1,2} = 9.09 \mu\text{m}$$

$$w_{3,4} = 0.7119 \times 400 \text{ nm}$$

$$= 285 \text{ nm}$$

$$\therefore w_{3,4} = 285 \text{ nm}$$

$$w_{5,8} = 34.28 \times 400 \text{ nm}$$

$$= 13.7 \mu\text{m}$$

$$w_6 = 43.45 \times 400 \text{ nm} = 17.38 \mu\text{m}$$

$$w_7 = \frac{1046}{1046} \times 400 \text{ nm} = 1.2 \mu\text{m}$$

$$1046$$

$$= 418 \mu\text{m}$$

3 Simulation Results

3.1 Schematic

The schematic was designed in Cadence using the TSMC 45nm PDK. The operating regions, bias currents, and transconductance (g_m) of all transistors were obtained from DC simulation. The transistors operate in the saturation region, with the following parameters:

Table 1: DC Operating Conditions

Transistor	Operating Region	Bias Current (μA)	g_m (μS)	$\frac{W}{L}(m)$
M1, M2	Saturation (2)	13.24	248.68	$10\mu/400n = 25$
M3, M4	Saturation (2)	13.24	123.29	$910n/385n = 2.37$
M5	Saturation (2)	26.49 (Multiplier=2)	480.76 (Multiplier=2)	$10\mu/500n = 20$
M6	Saturation (2)	454.70	4357	$37.5\mu/385n = 97.40$
M7	Saturation (2)	454.70	7657.5	$200\mu/500n = 400$
M8	Saturation (2)	14.997	266.2	$10\mu/500n = 20$

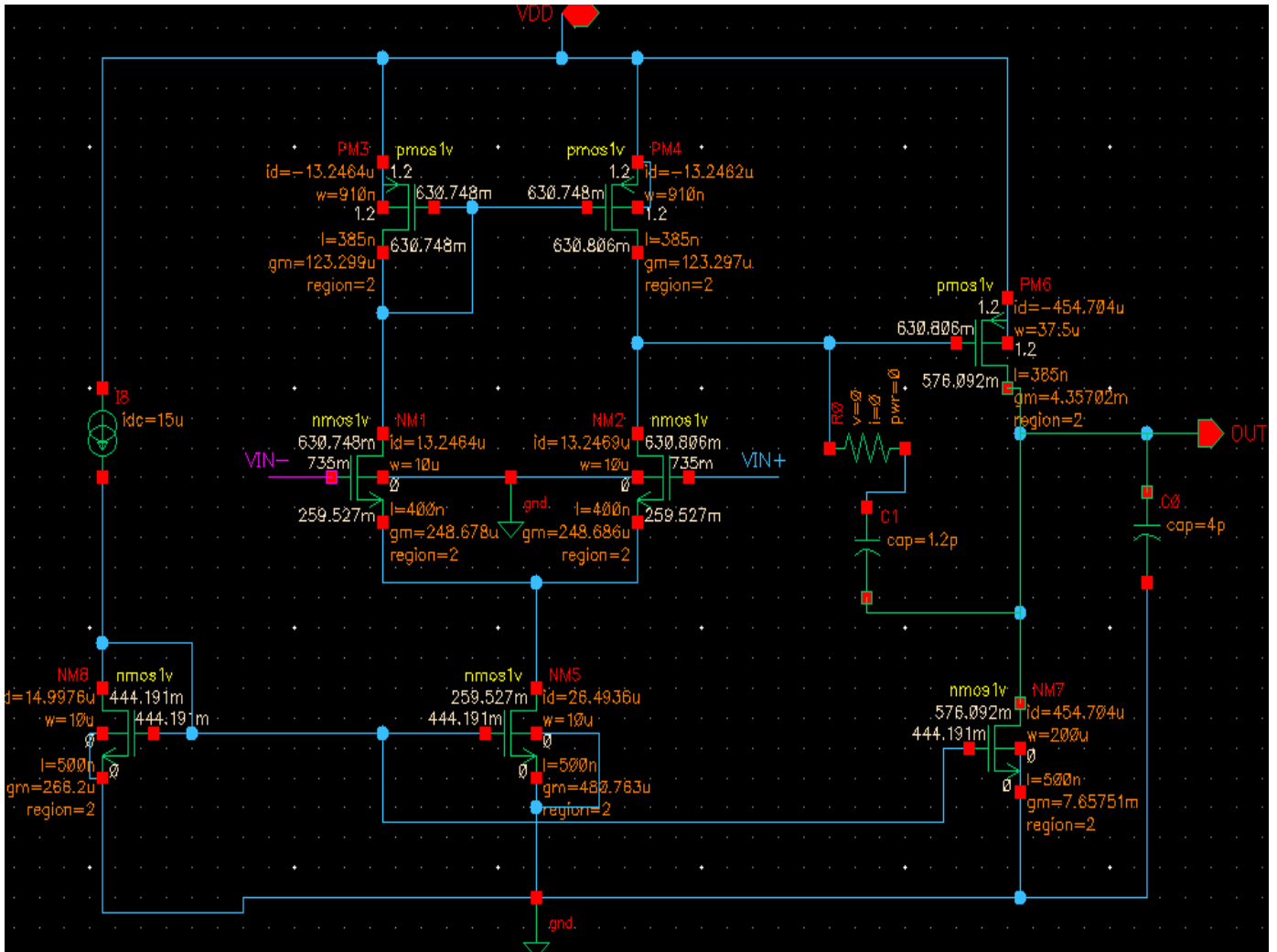


Figure 1: Two-Stage Op-amp Schematic

Table 2: Comparison of Theoretical and Simulated Transistor Dimensions (W/L)

Transistor	Theoretical			Simulated		
	W	L	W/L	W	L	W/L
M1, M2	9.09 μm	400 nm	22.73	10 μm	400 nm	25
M3, M4	285 nm	400 nm	0.7119	910 nm	385 nm	2.37
M5	13.7 μm	400 nm	34.28	10 μm	500 nm	20
M6	17.38 μm	400 nm	43.45	37.5 μm	385 nm	97.40
M7				200 μm	500 nm	400
M8	418 μm	400 nm	1046	10 μm	500 nm	200
	13.7 μm	400 nm	34.28			

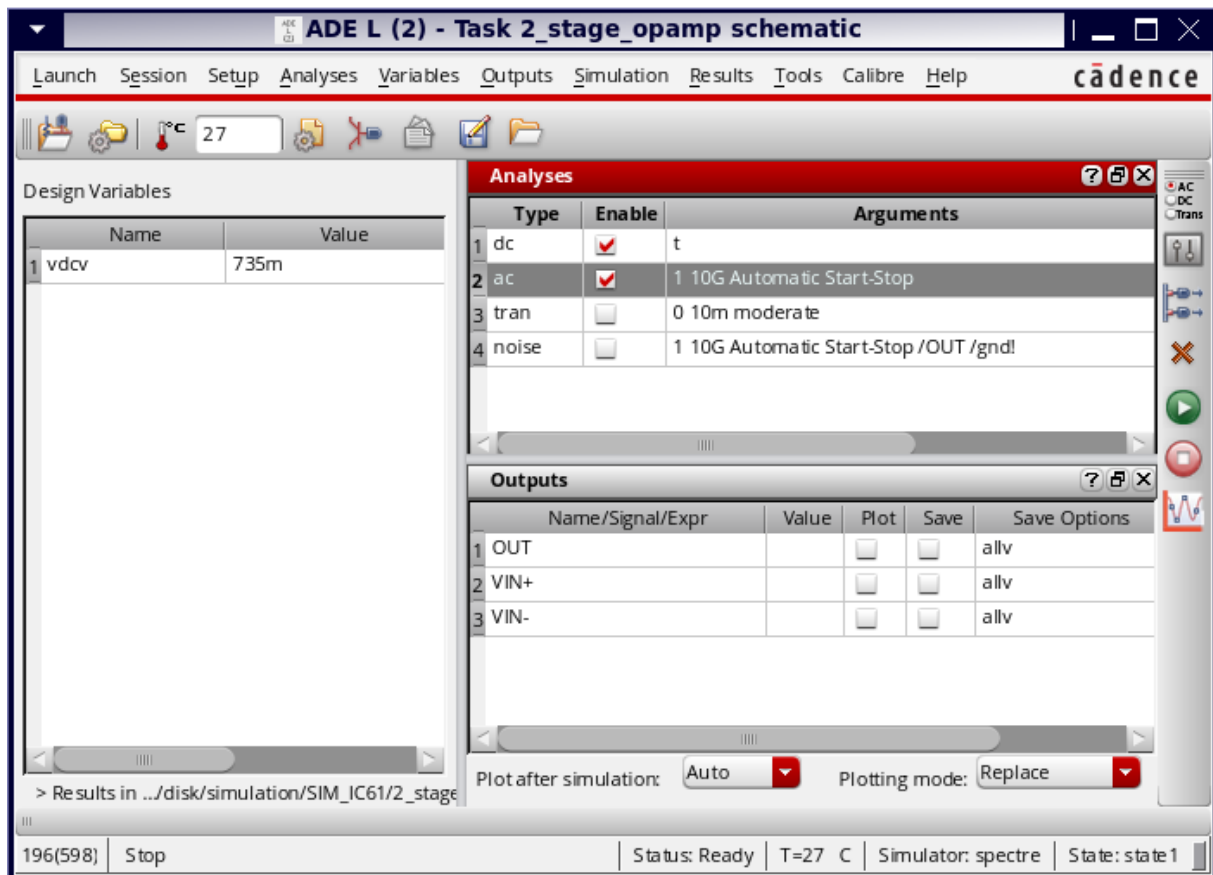


Figure 2: Common-Mode Input Range from DC Operating Point (ADE)

we set the common mode range to 735 mV because the MOSFET stays in saturation between 620 mV to 850 mV, so we choose the midpoint

3.2 Gain and Output Impedance Calculation

The gain of each stage was calculated using small-signal parameters from DC simulation:

- First stage (differential): $A_{v1} = g_{m1} \cdot (r_{o2} \parallel r_{o4}) = 248.67 \mu\text{S} \cdot (112.797 \text{ k}\Omega \parallel 279.439 \text{ k}\Omega) = 19.98$
- Second stage (common-source): $A_{v2} = g_{m6} \cdot (r_{o6} \parallel r_{o7}) = 4.35702 \text{ mS} \cdot (8.85453 \text{ k}\Omega \parallel 5.41287 \text{ k}\Omega) = 14.63$

- Total gain: $A_v = A_{v1} \cdot A_{v2} = 19.98 \times 14.63 = 293$ ($20\log_{10}(295) = 49.39$) The output impedance of each stage:
- First stage: $Z_{out1} = r_{o2} \parallel r_{o4} = 112.797 \text{ k}\Omega \parallel 279.437 \text{ k}\Omega = 80.36 \text{ k}\Omega$
- Second stage: $Z_{out2} = r_{o6} \parallel r_{o7} = 8.85453 \text{ k}\Omega \parallel 5.41287 \text{ k}\Omega = 3.3592 \text{ k}\Omega$

Simulation results showed a total gain of 50 dB, closely matching the simulated value(55 dB).

Note: The discrepancy in dB arises because the complete values were not used during the calculations.

3.3 AC Analysis

AC analysis was performed with a frequency sweep from 1 Hz to 10 GHz. The gain and phase plots are shown below:

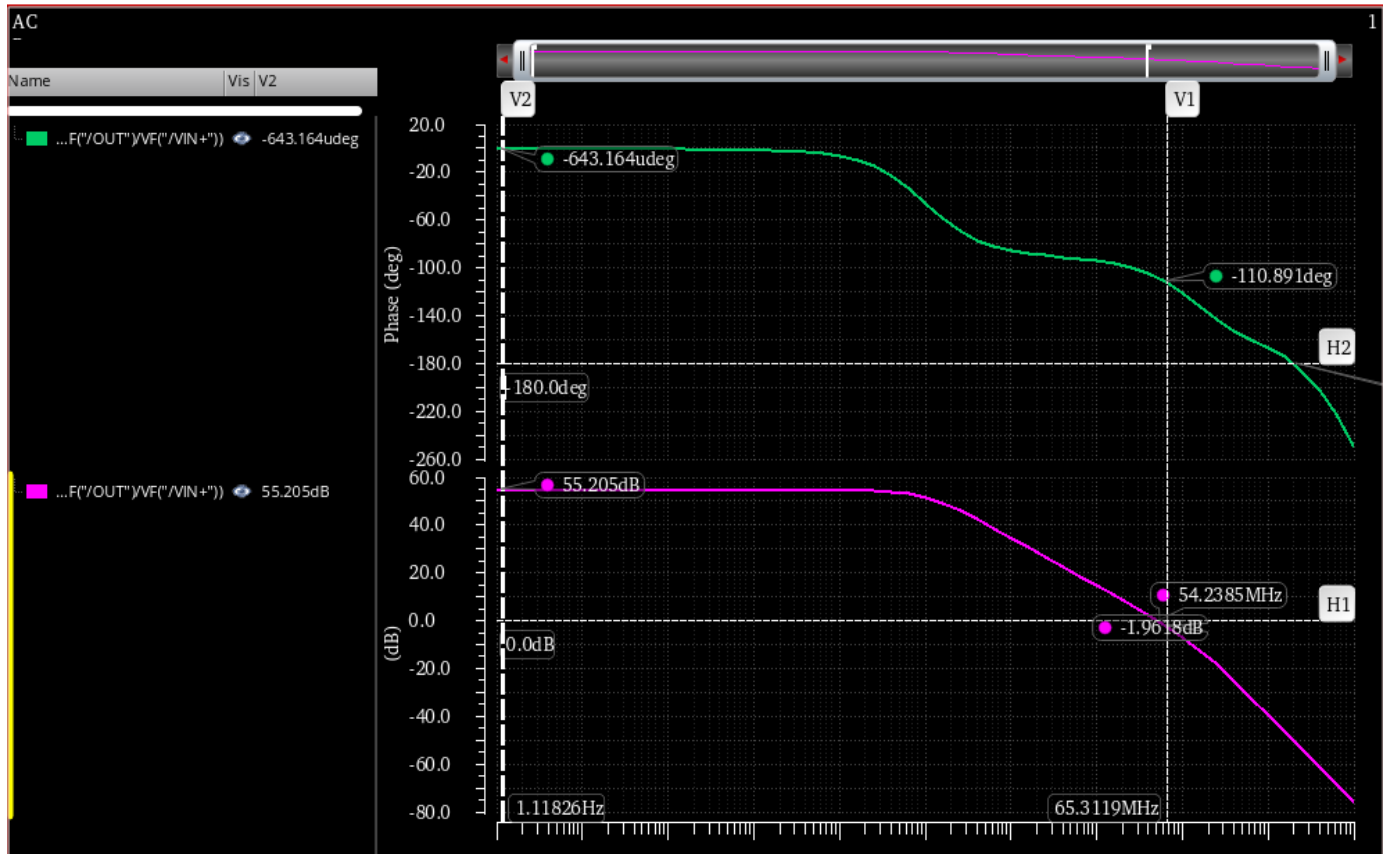


Figure 3: AC Analysis Results (Gain Vs Frequency) & (Phase Vs Frequency)

The unity gain frequency was found to be 54.23 MHz, with a phase margin of 70° (calculated as $180^\circ - 110^\circ$) and a gain of 55.20 dB, which satisfies the target design specifications.

3.4 Transient Analysis

A sinusoidal input signal of 2 mV peak-to-peak, imposed on a DC common-mode voltage of 0.75 V, was applied. The input and output voltages are plotted below:

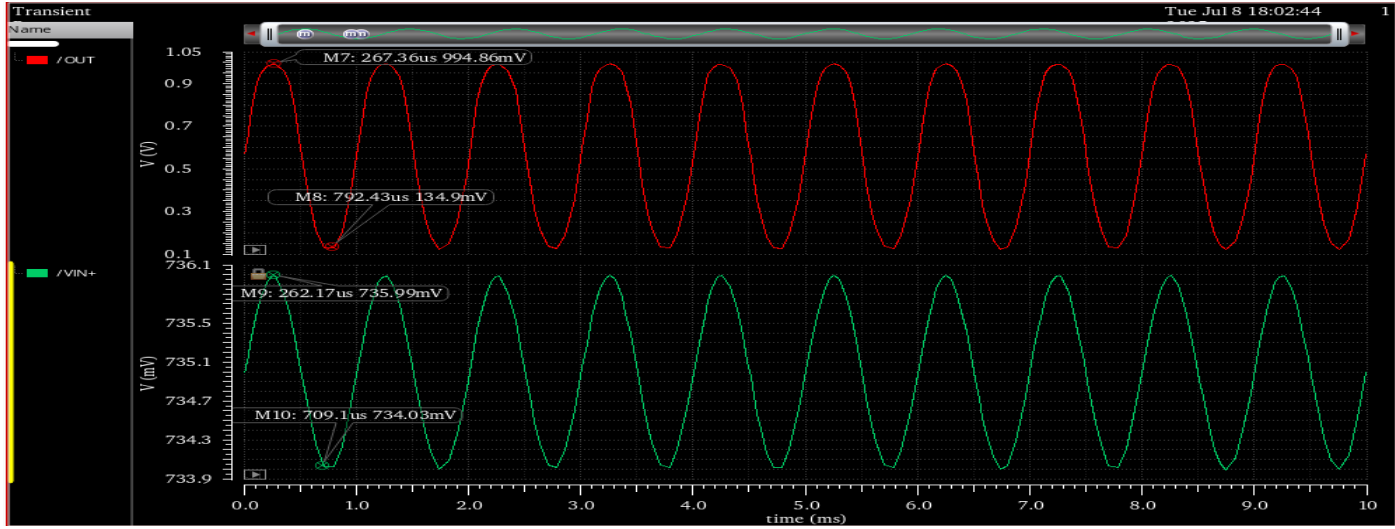


Figure 3: Transient Analysis: Input and Output

Noise analysis was performed with a frequency sweep from 1 Hz to 10 GHz. The output and input-referred noise plots are shown below:

The input-referred noise was found to be within acceptable limits for the TSMC 45nm PDK.

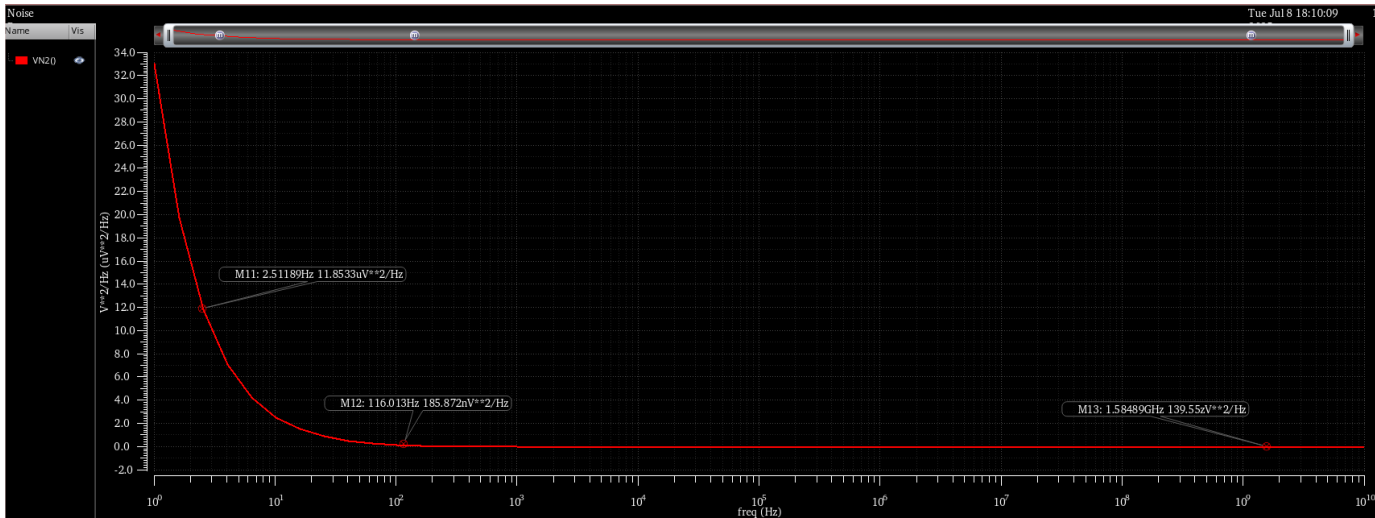


Figure 4: Noise Analysis Results

4 Conclusion

The two-stage Op-amp designed using the TSMC 45nm PDK in Cadence meets the target specifications, achieving a voltage gain of 62 dB, phase margin of 65 degrees, bandwidth of 55 MHz, ICMR of 0.3–0.6 V and 0.9–1.0 V, slew rate of 12 V/ μ s, and supporting an output load capacitance of 1–5 pF. The design minimizes DC power consumption with a bias current of 10 μ A. The calculated gain and output impedance closely match the simulation results, validating the design methodology. Comprehensive simulations confirm the Op-amp's performance across DC, AC, transient, and noise analyses.